ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** force acting per unit area	P = F / A.
2	**(a)** barometer	Barometer.
3	**(a)** very low pressure	Low pressure centre.
4	**(b)** pascal	1 Pa = 1 N/m squared.
5	**(a)** wind	Wind is air in motion.
6	**(c)** eye	Clear and calm.
7	**(c)** 50 Pa	100 / 2 = 50 Pa.
8	**(b)** differences in air pressure due to uneven heating	Air flows from high to low pressure.
9	**(a)** warm oceans	Needs warm water.
10	**(c)** its edge has a smaller area, so pressure is greater	Same force over smaller area gives larger pressure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force per unit area; P = F/A (1); SI unit pascal (Pa) (1).	1+1
12	Pressure exerted by the weight of the air (1); Acts on everything on the Earth's surface (1).	1+1
13	Calm, clear region at the centre (1); Surrounded by strong winds (1).	1+1
14	Small area gives large pressure (1); Easily enters the wood (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	P = F / A (1); = 600 / 0.5 (1); = 1200 Pa (1).	1+1+1
16	More air particles collide with the walls (1); Gas pressure increases inside (1); Balloon stretches (1).	1+1+1
17	Sea breeze blows towards land (1); Land air rises, cooler air flows in (1); Pressure difference creates the wind (1).	1+1+1
	OR: Sharp edge has small area (1); Same force gives larger pressure (1); Blunt edge spreads force over larger area (1).	1+1+1
18	Warm sea heats air (1); Warm moist air rises, low pressure forms (1); Surrounding air rushes in and spirals (1).	1+1+1
	OR: Gas particles move freely and randomly (1); They collide with walls (1); Collisions create pressure (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	P = F/A (1); Larger force or smaller area gives greater pressure (1); 50/0.25 = 200 Pa (1); 50/0.5 = 100 Pa (1); Examples: knife, camel's feet (1).	1+1+1+1+1
	OR: Measures air pressure (1); Falls with altitude (1); Low pressure signals storms (1); High pressure signals clear weather (1); Used in weather forecast (1).	1+1+1+1+1
20	Bottle with holes at different heights (1); Lowest stream goes farthest (1); Pressure increases with depth (1); A rubber-covered funnel in water shows pressure in all directions (1); Dams are thicker at the base (1).	1+1+1+1+1
	OR: Before: store water and food (1); Before: secure house (1); During: stay inside or in shelter (1); After: avoid fallen wires (1); Follow official instructions (1).	1+1+1+1+1

Mistakes to avoid:

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.

- Confusing pressure (N/m squared, pascal) with force (N, newton).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** rises up	It expands and becomes lighter.
2	**(d)** hurricanes	Names depend on region.
3	**(d)** reduce pressure on the shoulders	Larger area, less pressure.
4	**(b)** high pressure to low pressure	To equalise pressure.
5	**(a)** typhoons	Regional name.
6	**(c)** depth	More liquid above.
7	**(a)** the sea to the land	Land heats faster, air rises over land.
8	**(c)** a narrow funnel-shaped rotating column of air	Violent whirling storm.
9	**(d)** in all directions	Liquid pressure acts in all directions.
10	**(d)** night	Land cools faster than sea.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Outside pressure falls with height (1); Pressure inside the ear is higher (1).	1+1
12	Whirling funnel of air from cloud to ground (1); Very destructive (1).	1+1
13	Large area reduces pressure (1); Vehicle does not sink in soft ground (1).	1+1
14	Uneven heating of the Earth's surface (1); Air moves from high to low pressure (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Make holes at different heights in a bottle (1); Fill with water (1); Water from the lowest hole goes farthest (1).	1+1+1
16	Heavy rain and floods (1); Strong winds destroy buildings (1); Storm surge on the coast (1).	1+1+1
17	Pressure of water increases at the bottom (1); Broad base withstands it (1); Prevents collapse (1).	1+1+1
	OR: Water pressure increases with depth (1); Normal body cannot bear the pressure (1); Suit protects the diver (1).	1+1+1
18	Move to shelters (1); Switch off electricity (1); Keep radio and torch ready (1).	1+1+1
	OR: Early alert helps evacuate (1); Fishermen avoid going to sea (1); Services get ready (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Air has weight and exerts pressure (1); Suction cup (1); Drinking through a straw (1); A card on an inverted glass of water (1); A collapsing can (1).	1+1+1+1+1
	OR: Particle collisions create pressure (1); Smaller volume means more collisions, higher pressure (1); Heating increases speed and collisions (1); Pressure rises with temperature (1); Examples: tyre, pressure cooker (1).	1+1+1+1+1
20	Uneven heating (1); Warm air rises, low pressure (1); Cool air from high pressure moves in (1); Sea breeze by day (1); Land breeze by night (1).	1+1+1+1+1
	OR: Knives and needles: small area (1); Tractor wheels: wide (1); Foundations: broad (1); Camel feet (1); Skis (1).	1+1+1+1+1

Mistakes to avoid:

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- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** lightning and thunder	Electric discharge in clouds.
2	**(c)** water pressure increases with depth	Greater pressure at depth.
3	**(c)** atmospheric pressure	Pressure outside pushes it.
4	**(c)** stay indoors and listen to official updates	Safety.
5	**(d)** atmospheric pressure	Air has weight.
6	**(d)** atmospheric pressure pushes the liquid up	Pressure inside the mouth falls.
7	**(a)** to protect people from storms and floods	Safe place.
8	**(b)** lower	Less air above.
9	**(c)** collisions of gas particles with the walls	Particle collisions.
10	**(d)** anemometer	Anemometer.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Barometer: pressure (1); Anemometer: wind speed (1).	1+1
12	Increases with depth (1); Acts in all directions (1).	1+1
13	Sea breeze blows from sea to land during the day (1); Land breeze blows from land to sea at night (1).	1+1
14	Stay indoors (1); Avoid tall trees and water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Measures air pressure (1); Reading falls with height (1); Less air above (1).	1+1+1
16	Warm moist air rises quickly (1); Clouds develop with charges (1); Lightning and thunder with heavy rain (1).	1+1+1
17	Sucking lowers pressure inside the straw (1); Atmospheric pressure pushes the liquid up (1); Liquid rises into the mouth (1).	1+1+1
	OR: Land heats faster in the day, warm air rises (1); Cooler sea air moves in as sea breeze (1); At night, reverse as land breeze (1).	1+1+1
18	Heated air expands and becomes lighter (1); Surrounding denser air pushes it up (1); It floats upward (1).	1+1+1
	OR: Storm surge and flooding (1); Damage to houses and crops (1); Power and communication loss (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Eye with calm weather (1); Eye wall with strong winds (1); Forms over warm oceans (1); Warm moist air supplies energy (1); On land the supply is cut off (1).	1+1+1+1+1
	OR: Day: land warmer (1); Air over land rises (1); Low pressure over land (1); Cool air from sea moves in (1); Night reverses (1).	1+1+1+1+1
20	Thunderstorm: lightning, rain (1); Tornado: destructive spin (1); Cyclone: storm surge and floods (1); Warnings and shelters (1); Education and drills (1).	1+1+1+1+1
	OR: Fishermen avoid sea (1); Farmers protect crops (1); Evacuation planned (1); Alerts by colour codes (1); Reduces losses (1).	1+1+1+1+1

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